

NLP intro

Natural Language Processing for the Digital Humanities

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What is NLP?

Natural Language Processing (NLP) is a field of Artificial Intelligence that gives the machines the ability to read, understand, and derive meaning from human languages.

NLP is a discipline where *computer science, artificial intelligence, and cognitive logic are intercepted*, with the objective to read, decipher, understand, and make sense of the human language in a valuable way.

Connected fields

NLP is often closely related to information retrieval and computational linguistics (Eisenstein, 2019).

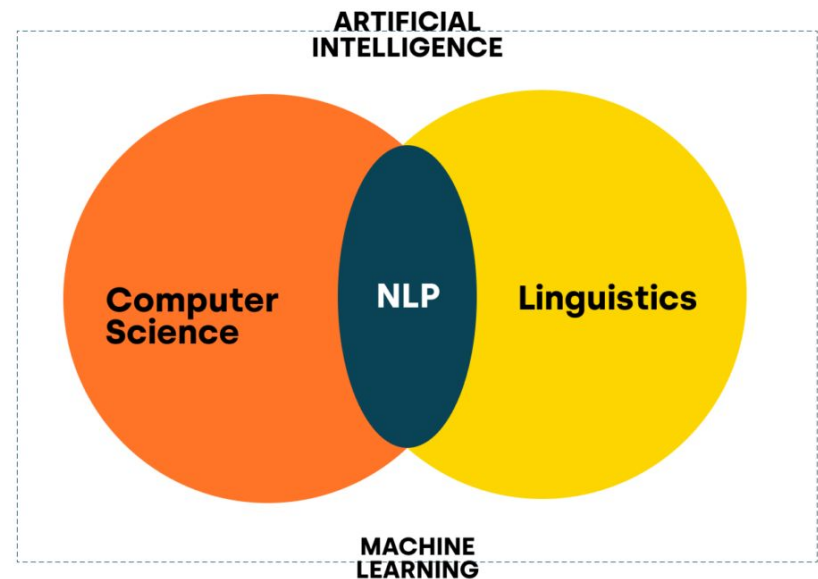
It includes several main subfields:

- natural language understanding
- speech recognition
- natural language generation
- text classification

NLP and language

... helps computers understand, interpret and produce human language.

... studies language as data and develops models that can analyse linguistic structure, meaning and context in both written and spoken communication.



<https://www.qtravel.ai/blog/what-is-natural-language-processing-and-how-do-chatgpt-systems-work/>

What makes language difficult
for computers?

can you think of some
examples?

Lexical ambiguity

- a single word can have multiple meanings in a singular language, depending on the context within a sentence
- word-sense disambiguation with POS tagging, nowadays models that learn context in both directions (e.g. transformers)



<https://www.thoughtco.com/ambiguity-language-1692388>

Syntactic and semantic ambiguity

- the sentence allows for multiple interpretations, depending on how it is put together
- understanding still based on context of text surrounding the sentence



<https://www.thoughtco.com/ambiguity-language-1692388>

Pragmatic ambiguity

- meaning depends on tone, intent or situation / social context
- very common in everyday conversations
- sarcasm, irony



"I must be pretty smart.
My teacher said I'm a real wise guy."

https://www.cartoonstock.com/directory/t/teacher_s_sarcasm.asp?srsid=AU7gw4VRZ9rC1XQZLq2wP6gs2hKZAY/cdt6b4OEvb_n76ibQtpqKkGB00

Referential ambiguity

- when a pronoun (like "he," "she," "it," or "they") or a phrase is unclear about what or who it is referring to.

“Alice told Jane she would win the prize.”

Ellipsis ambiguity

- when a part of the sentence is left out

“I like my dog more than my sister.”

Language becomes data - common NLP tasks

One of the most exciting aspects of NLP is the extraction of "meaning" from text, and a lot of NLP techniques revolve around this idea.

Which NLP techniques / tasks have you heard of?

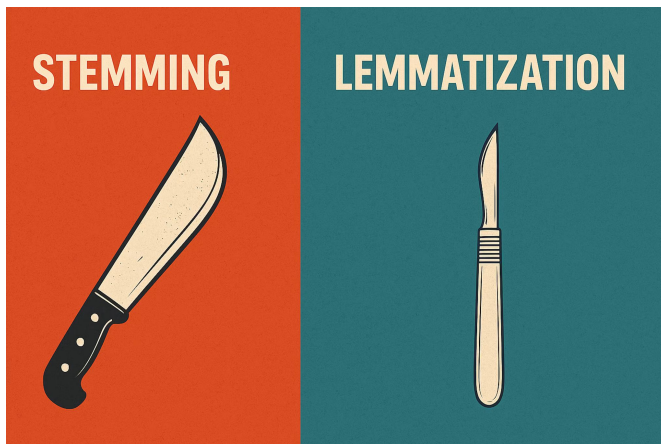
NLP tasks

Morphological:

- POS tagging

The/DT day/NN was/VBD extremely/RB rainy/JJ and/CC windy/NN and/CC that/IN made/NN me/PRP feel/VBP like/IN good/JJ things/NNS were/VBD bound/VBN to/TO happen/VB ./.

- lemmatization (was - to be)
- stemming (ability - abili, developing - develop)

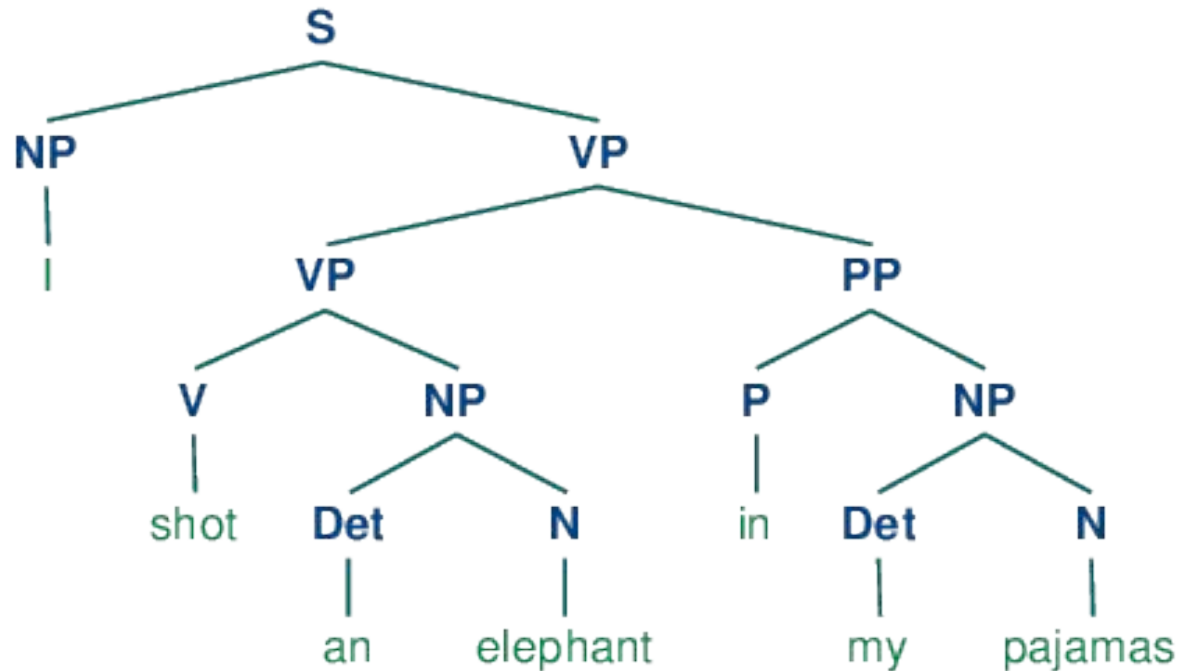


<https://hackernoon.com/stemming-vs-lemmatization-what-healthcare-text-data-taught-me-about-nlp-choices>

NLP tasks

Syntactic:

- tokenization (word or sentence based splitting)
- parsing



NLP tasks

Syntactic, pragmatic and semantic

- Named Entity Recognition (NER)
- Sentiment Analysis, Emotion Detection, Opinion Mining
- Topic Modeling
- Automatic Summarization
- Optical Character Recognition (OCR)
- Text generation
- Question Answering
- ...

Speech:

- speech processing, recognition, segmentation
- text-to-speech, speech-to-text

NLP tasks example

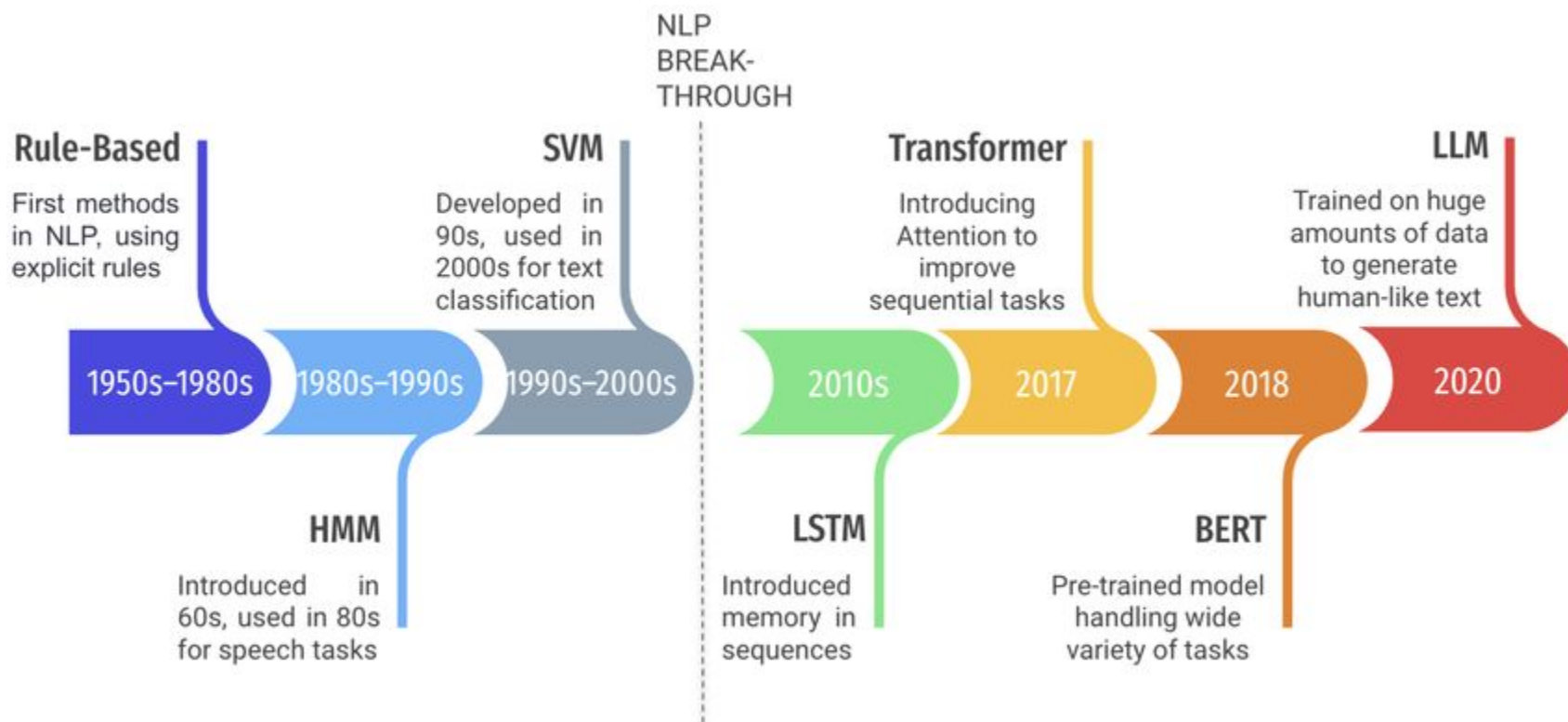
“Harry Potter is happy with the his new broom.”

An NLP system can, e.g.:

- Detect Harry Potter as a person (Named Entity Recognition)
- Identify broom as an object (Part Of Speech Tagging)
- Recognize sentiment as positive (Sentiment Analysis)
- Understand topic as product review (Topic Modelling)



NLP method progression



(Kra

sadakis et al., 2024)

My own path

1. Bachelor thesis (2017)
“Comparative analysis of stemmers, lemmatizers and POS taggers for English language” (<https://zir.nsk.hr/object/ffri:2119/FILE0>)
2. Master thesis (2019)
“Extractive summarization of scientific papers”
(<https://repository.ffri.uniri.hr/object/ffri:2136>)
3. PhD thesis (2026)
“Attitudes towards migration and minority groups in Austrian historical newspapers”
(<https://unipub.uni-graz.at/obvugrhs/content/titleinfo/13596677>):
topic modeling, BERT, sentiment analysis, corpus building

NLP applications in industry and science

NLP Applications

- 01 Voice Assistants
- 02 Text Analysis
- 03 Stopword Removal
- 04 Speech-to-Text
- 05 Chatbots
- 06 Content Recommendation

<https://www.geeksforgeeks.org/nlp/introduction-to-natural-language-processing-nlp/>

Why NLP for Digital Humanities?

- Investigate thousands or millions of texts rather than a small selection
- Discover patterns: identify recurring words, themes, entities, emotions, linguistic structures, and relationships that may be difficult to notice manually.
- Compare collections: authors, genres, historical periods, languages, communities, or types of discourse.
- Trace change over time: investigate changes in vocabulary, meaning, sentiment, topics, and discourse.
- Structure unstructured sources: extract people, places, dates, events, relationships, and other information from textual collections.

Why NLP for Digital Humanities?

- Search by meaning, not only words: embeddings and semantic search allow researchers to find conceptually related passages even when different vocabulary is used.
- Study large cultural collections: newspapers, books, parliamentary debates, correspondence, historical archives, etc
- Ask new research questions: computational methods make some questions feasible that would otherwise be practically impossible to investigate

What can we study?

- authorship, themes, genres, sentiment/emotion, entities, discourse, semantic change, historical language, multilingual corpora, low-resource languages, historical sources, character analysis, linguistic creativity, development of philosophical ideas, public discourse, stereotypes...

<https://www.nlp4dh.com/>

Alfred Escher an Heinrich Schweizer, s.l., Samstag / Sonntag / Montag, s.d.

Schlagwörter: Berufsleben, Freundschaften, Privatunterricht

Eidierter Text **Diplomatischer Text und Bild** Bild und diplomatischer Text Digitalisat

Samstag Abend, 8 Uhr.

In meinem Arbeitszimmer ganz einsam!

Mein theuerster Freund!

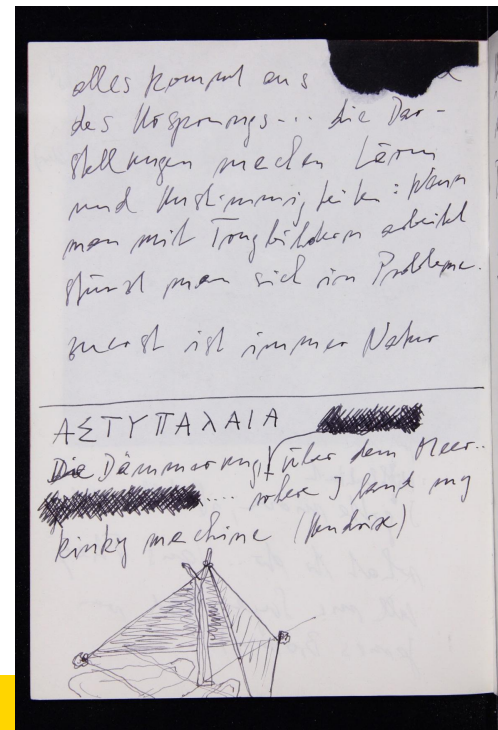
Von Neuem haben Sie uns verlassen! Wie weh thut es doch, sich von einem so lieben, guten Freunde trennen zu müssen! Nicht wahr, Sie lassen erfreuen uns bald wieder durch Ihre Ankunft, die uns immer die größte Freude verur-

gottvertrauen! Etwa 700 bis 800 Personen versammelten sich zum Gottesdienst.

Sontag Morgens ½ 10 Uhr.

Ich habe eben wieder den Brief vom 17^{ten} *huj.* gelesen, den ich an dem Montage erhielt, an welchem Sie zum ersten Mahle in *Belvoir* schliefen. Ich kann nicht Anders, als mich noch ein wenig mit Ihnen "darüber" unterhalten:

Sie sagen, theurer Lehrer, daß es Ihnen so wehe thue, von mir deßwegen Vorwürfe zu hören, weil Sie vorletzte Woche nicht nach Zürich kommen konnten. Der Ausdruck «Vor-



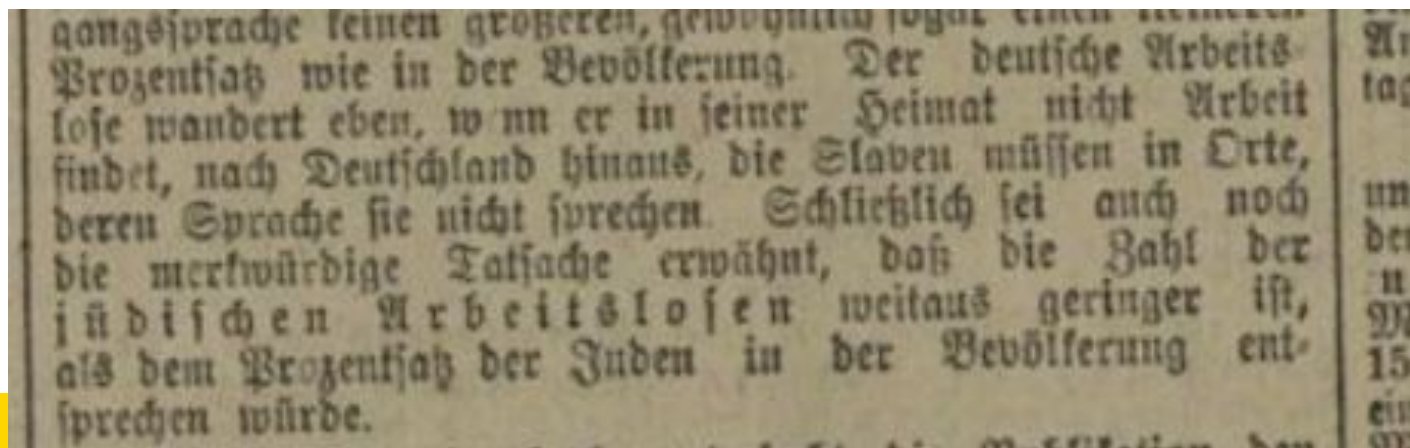
NLP challenges

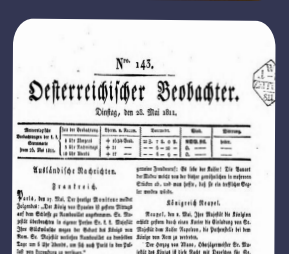
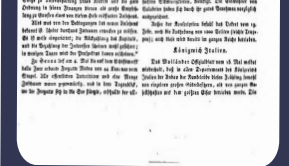
- data collection, annotation
- bias detection
- explainability
- safety and security
- efficient
- fair representation learning
- multilingualism
- evaluation of models...



NLP challenges in the DH

- historical language: spelling, grammar, meaning and language change
- OCR / HTR errors
- non-standard language
- metaphors, satire, irony, sarcasm, poetic language
- small datasets, low-resource languages
- complex data
- domain mismatch
- unstructured data





Attitudes towards migration and minority groups in Austrian historical newspapers

Lucija Brozić

University of Graz, Department of Digital Humanities



Motivation

MIGRATION & NEWSPAPERS

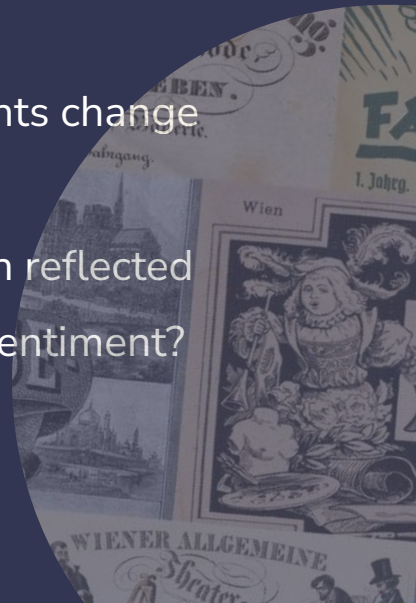
A central societal
phenomenon shaped by
political, economic, and social
change
(Steidl, 2021)

Modern media narratives
present past migration as
"good" and today's migration
as "problematic"
(Schrover, 2022)

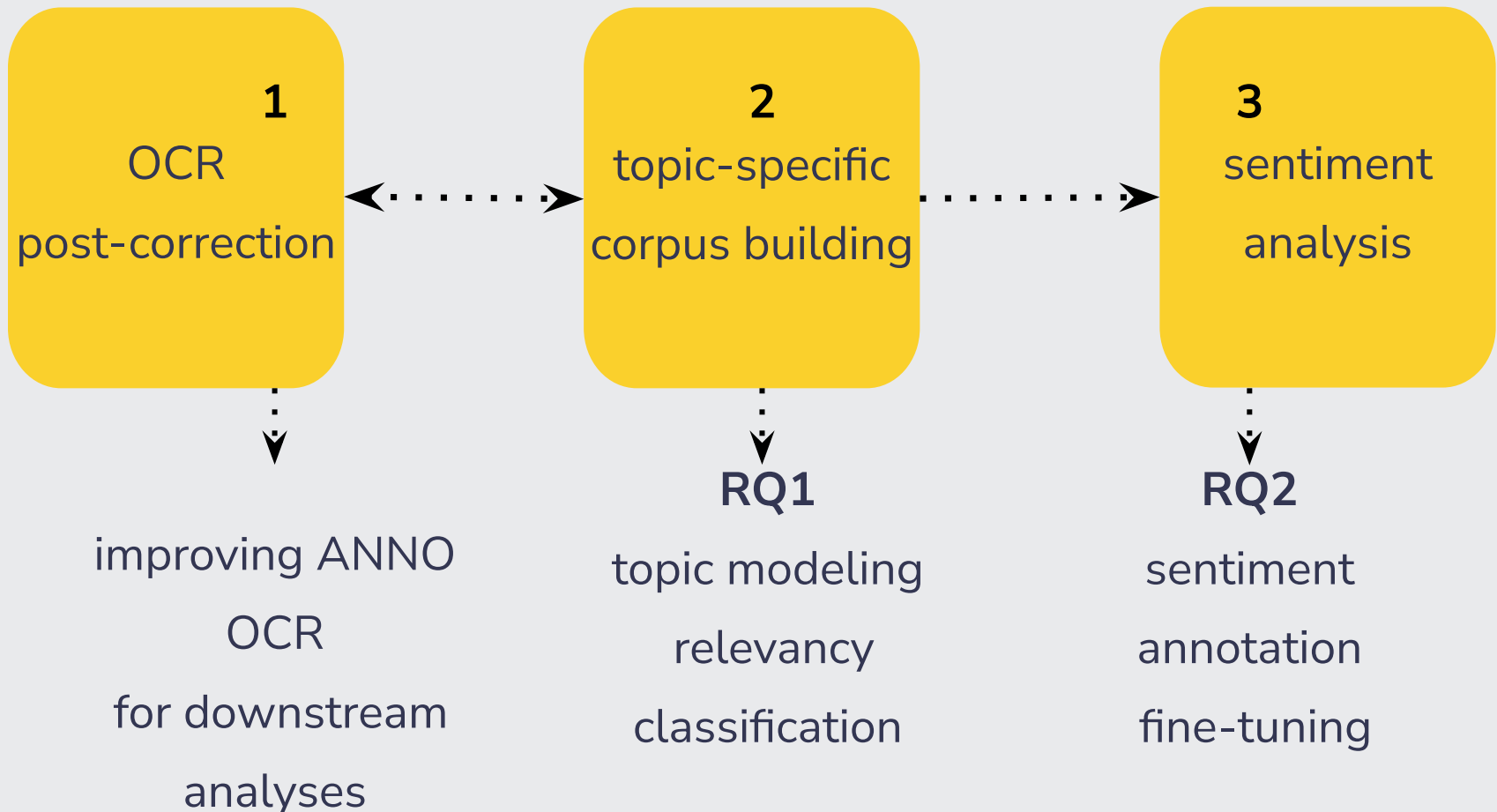
RESEARCH QUESTIONS

**How are migrants and
minorities represented in
newspaper discourse?**

1. What was the predominant sentiment?
2. How do the sentiments change over time?
3. Is political orientation reflected in the newspaper's sentiment?



Methodology and results



Conclusions

RQ1: How do topics related to migrants and minorities appear in the newspapers?

- frequency and content of migration and minority topics changes over time, consistent with historical events and processes

RQ2: How are migrants and minorities represented in newspaper discourse?

- polarity and sentiments influenced by date of publishing and political orientation

Limitations

- newspapers offer a singular perspective (Daut & Gabriel, 2023)
- bias introduced at every step of methodology
- documentation a must

topic-specific corpus MigraAnno

sentiment gold-standard corpus SentiAnno

ML models for OCR post-correction, topic modeling, relevancy classification and sentiment analysis



Exercise 0

Think of your own Bachelor or Master thesis (or any other project you would like to work on).

Which research questions can be more easily answered with NLP techniques?

Which techniques would you use?

How can you advance NLP methods themselves through your research?

Share with the class.